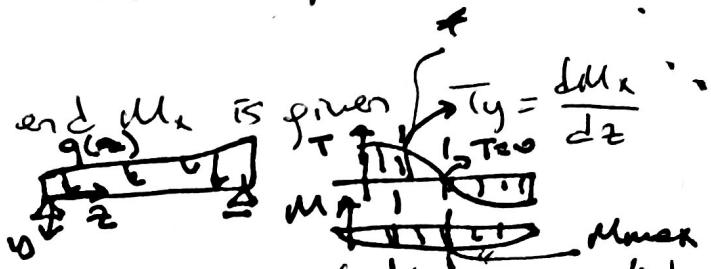
BENDING WITH SHEAR 1 week

In the case of shear with bending, shear force and bending moment act together

The relation between T_y and M_x is given $T_y = \frac{dM_x}{dz}$

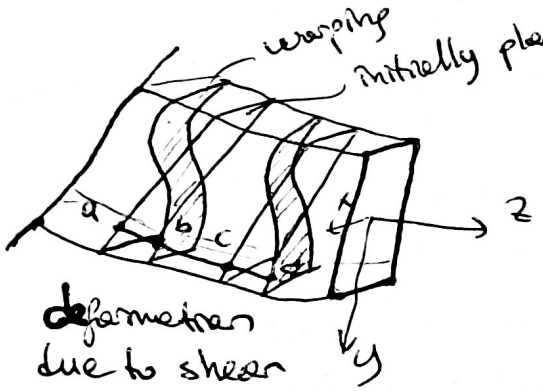


Calculation of Normal stress:

If $T_y = 0$ "the plane sections remain plane after bending is valid

Hence $\sigma_z = \frac{M_x}{I_x} y$ can be used to calculate normal stresses

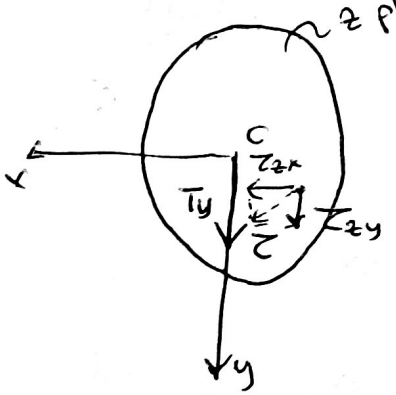
If $T_y \neq 0$, then, the sections are warped due to the shear force



Since the distances between ac and bd equal ($ac = bd$) Then, the formula (The length of fibers does not change due to shear) $\sigma_z = \frac{M_x}{I_x} \cdot y$ can still be used to calculate normal stress.

(Normal stress distribution is still assumed to be linear along the cross-section height)

Calculation of Shear Stresses:



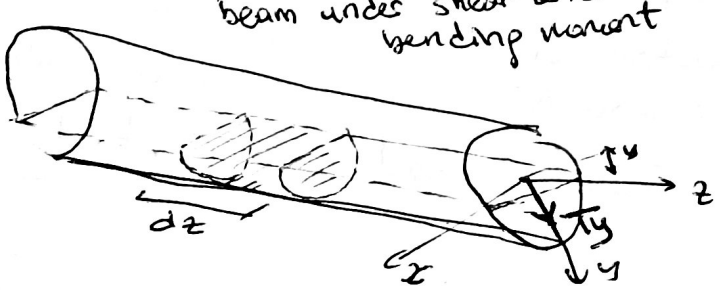
Using equilibrium conditions, it can be written

$$\iint \tau_{zx}\, dA = 0 \qquad \iint \tau_{zy}\, dA = T_y$$

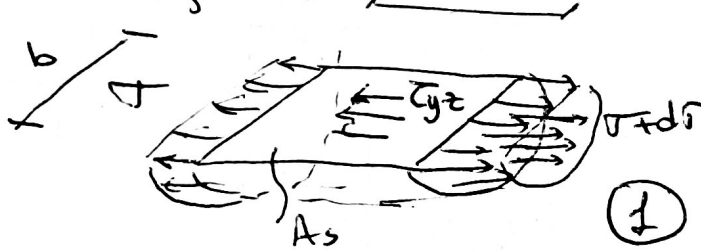
These equations are not enough to fully define the shear stress distribution, additional equations are needed.

Calculation of τ_{zy} when $T_y > 0$ and $T_x = 0$

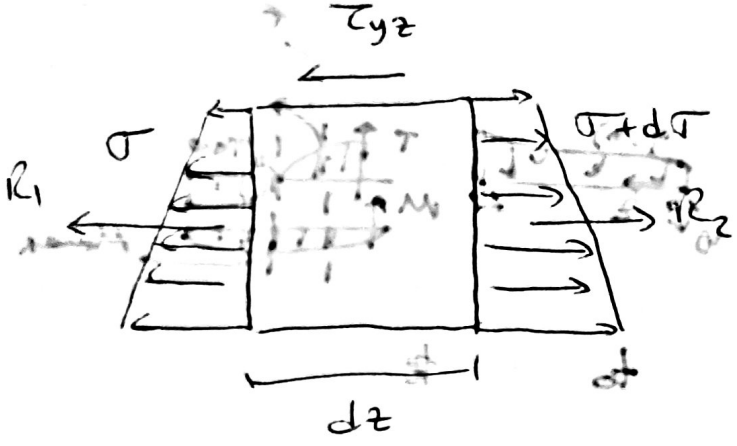
beam under shear and bending moment



extract the portion with length dz



Perpective view (2D)



R_1, R_2 are the stress resultants

$$R_1 = \int_{A_s} \sigma dA = \int \frac{M_x}{I_x} y dA$$

$$= \frac{M_x}{I_x} \int_{A_s} y dA = \frac{M_x}{I_x} S_x$$

$$R_2 = \int_{A_s} (\sigma + d\sigma) dA = \int_{A_s} \frac{M_x + dM_x}{I_x} y \, dA$$

$$= \frac{M_x + dM_x}{I_x} S_x$$

S_x is the static moment

from the equilibrium condition of the lower part

$$-\tau_{yz} b dz - R_1 + R_2 = 0 \rightarrow -\tau_{yz} b dz - \frac{M_x}{I_x} S_x + \frac{M_x + dM_x}{I_x} S_x = 0$$

$$\rightarrow \tau_{yz} = \frac{dM_x}{dz} \frac{S_x}{I_x} \cdot \frac{1}{b} = \frac{\tau_y S_x(y)}{I_x b}$$

$(y>0, S_x>0)$

if the shear stress is to calculate the upper part, Then

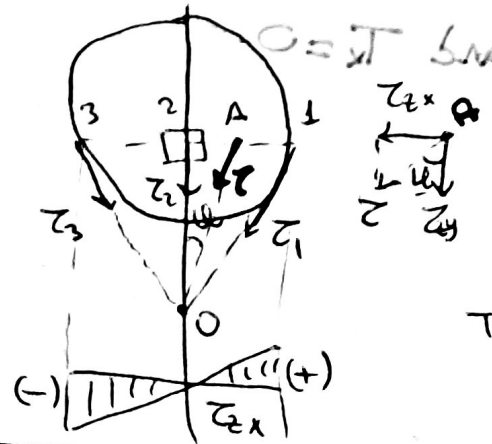
$$\tau_{zy} = -\frac{\tau_y S_x(y)}{I_x b} \quad (y<0, S_x<0)$$

(b is assumed to be constant)

If b is not constant $\tau_{zy} = \frac{\tau_y S_x(y)}{I_x b(y)}$

for max. shear stress $(\tau_{zy})_{max} = \frac{\tau_y}{I_x} \left(\frac{S_x(y)}{b(y)}\right)_{max}$

Calculation of τ_{zx} when $\tau_y > 0$:



If we know the direction of τ and τ_{zy} is calculated from the previous equation then, τ_{zx} can be calculated from vector algebra

After point A is determined, it is calculated

Then τ_{zx} is obtained $\tau_{zx} = \tau_{zy} \tan\phi$

Finally $\tau = \sqrt{\tau_{zx}^2 + \tau_{zy}^2}$

The relation between max shear stress τ_{max} and average shear stress τ_{avg} is given as

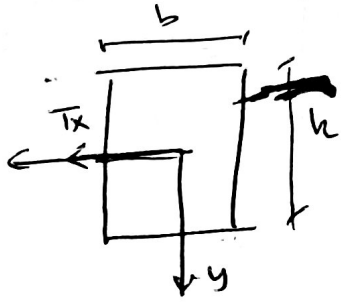
$$\tau_{max} = k_y \tau_{avg} \quad \text{where} \quad \tau_{avg} = \frac{T_y}{A}$$

k_y depends on the cross-section's geometry.

if $T_x > 0$ $T_y = 0$

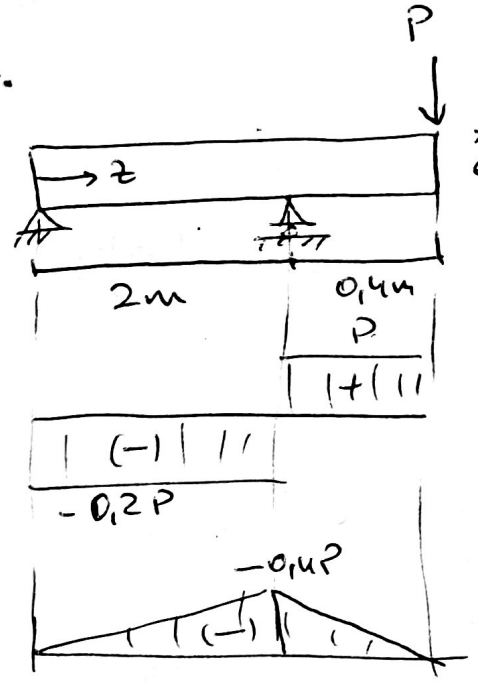
$$\tau_{zx} = \frac{S_y(x)\, T_x}{h(x)\, I_y} \quad (x>0, S_y>0)$$

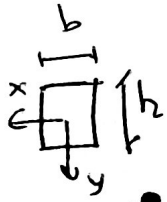
$$\tau_{zx} = -\frac{S_y(x)\, T_x}{h(x)\, I_y} \quad (x<0, S_y>0)$$



$I\left(e-\frac{c}{5}\right)$

Ex:





Calculate the load P

if $\tau_{em} = 10$ MPa and $\tau_{em} = 1{,}4$ MPa

$h = 80$ mm $b = 40$ mm

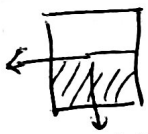
from (M) and (T) diagrams, it is seen that

$T_{max} = P$ and $M_{max} = -400P$ (N/mm)

$$I_x = \frac{bh^3}{12} = 2{,}43\,10^6 \text{ mm}^4$$

$$S_x = \left(\frac{1}{4}h\right)\left(\frac{1}{2}bh\right) = 40500 \text{ mm}^4$$

(Static moment of upper part of the cross section)



Safety check for bending

$$\tau_{max} = \frac{M_{max}}{I_x}\cdot\left(\frac{h}{2}\right) \le \tau_{em} = 10$$

$$P \le \frac{10 \cdot 2{,}43\,10^6}{-45 \cdot -400} = 1350 \text{ N}$$

Safety check for shear stress

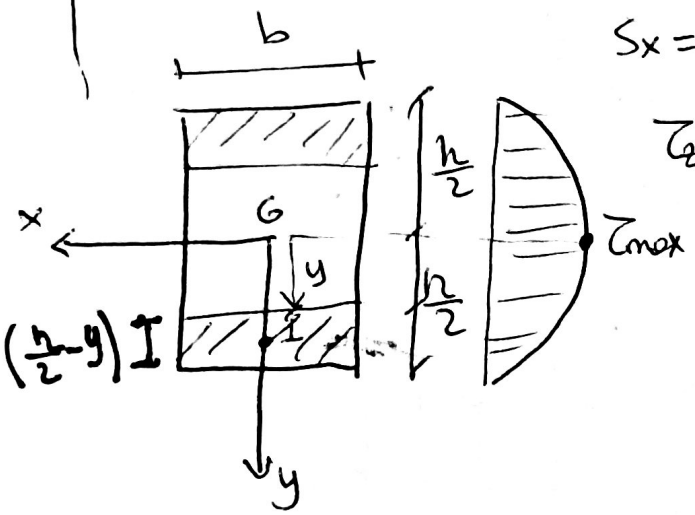
$$\tau_{max} = -\frac{T_{max}\, S_x}{I_x\, b} < \tau_{em} = 1.4$$

$$P \le -\frac{1.4\,(2{,}43\,10^6).40}{-40500} = 3360 \text{ N}$$

Hence max. $P = 1350$ N (smaller P is chosen)

Stress Distribution for various cross-section shapes

Rectangular Shape



$$S_x = b\left(\frac{h}{2}-y\right)\left(y+\left(\frac{h}{4}-\frac{y}{2}\right)\right) = \frac{1}{2}\left(\frac{h^2}{4}-y^2\right)$$

$$\tau_{zy} = \frac{T_y S_x}{I_x b} = \frac{T_y\left(\frac{1}{2}\left(\frac{h^2}{4}-y^2\right)\right)}{\frac{bh^3}{12}\, b}$$

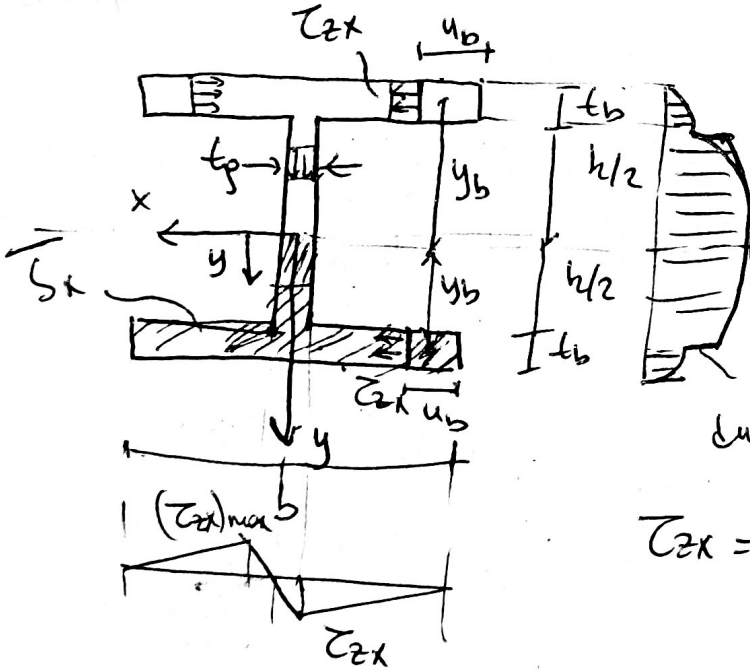
$$= \frac{6\,T_y}{bh^3}\left(\frac{h^2}{4}-y^2\right)$$

$$(\tau_{zy})_{max} = \tau_{zy}\Big|_{y=0} = \frac{3\,T_y}{2bh}$$

$$\tau_{avg} = \frac{T_y}{bh}, \text{ hence } (\tau_{zy})_{max} = \frac{3}{2}\tau_{avg}$$

$$k_y = \frac{3}{2}$$

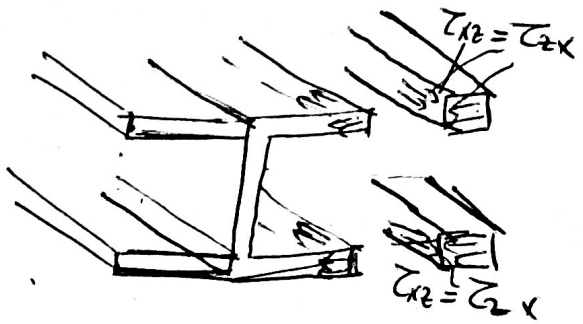
Profile cross-section



$$(\tau_{zy})_{max} = \frac{T_y \bar{S}_x}{I_x t_g}$$ → ~~loading~~ cross-section width

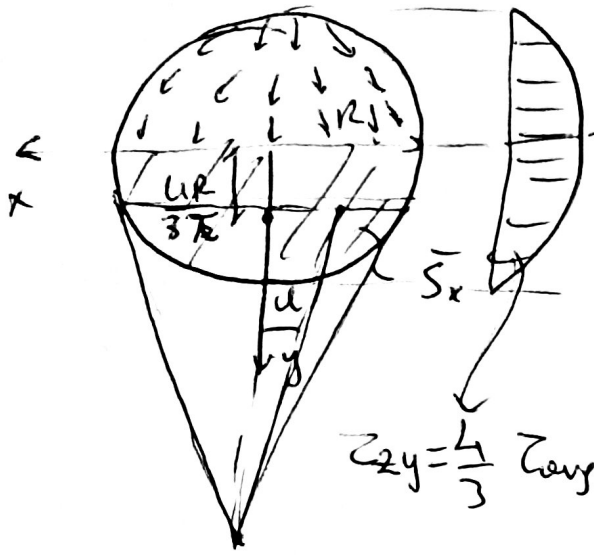
discontinuity due to change of width

$$\tau_{zx} = \frac{T_y (S_x)_b}{I_x t_b} \qquad (S_x)_b = u_b t_b \left(\frac{h}{2}+\frac{t_b}{2}\right)$$



Circular Cross-Section

$$\bar{S}_x = \frac{\pi R^2}{2} \cdot \frac{4R}{3\pi} = \frac{2}{3}R^3$$



$$\tau_{max} = \frac{T_y \bar{S}_x}{I_x 2R} = \frac{T_y \frac{2}{3}R^3}{\frac{\pi R^4}{4} 2R} = \frac{4T_y}{3\pi R^2}$$

$$\tau_{max} = k_y \tau_{avg}$$

$$\frac{4T_y}{3\pi R^2} = k_y \frac{T_y}{\pi R^2}$$

$$k_y = \frac{4}{3}$$

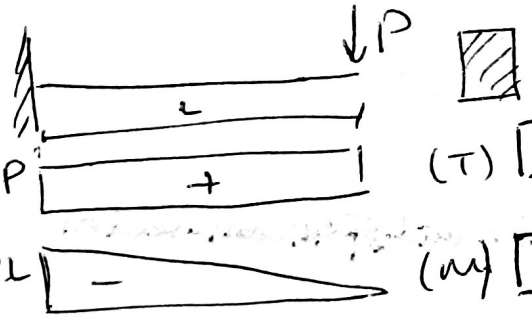
$$\tau_{zy} = \frac{4}{3}\tau_{avg}\left[1 - \frac{y^2}{R^2}\right]$$

$$\tau_{zx} = \underbrace{\left(\frac{-xy}{(R^2-y^2)}\right)}_{\text{tensile}} \tau_{zy}$$

Cross-Section Dimensioning

Normal and shear stresses cause two dimensional stress state on the points of cross-section

However σ_{max} and τ_{max} occur at different locations in general.



$$\sigma_{max} = \frac{M}{\frac{bh^3}{12}} \cdot \frac{h}{2} \quad \text{(occurs at the farthest fiber)}$$

$$= \frac{6PL}{bh^2}$$

$$\tau_{max} = k_y \tau_{avg} = \frac{3}{2}\frac{P}{bh} \quad \text{(occurs at } y = 0\text{)}$$

$$\frac{\tau_{max}}{\sigma_{max}} = \frac{h}{4L}$$

In general h/L is very small (incl.) Hence the important stress is σ_{max}

Thus the cross section is dimensioned according to σ_{max} and checked for τ_{max}. (e.g $k_y \frac{T}{A} \leq \tau_{em}$)

In some cases, cross section is checked for two-dimensional stress state arise from both τ and σ. In this case one of the failure theories (e.g. TRESCA, VON MISES) is used.

$$f(\sigma_1, \sigma_2) \leq \sigma_M$$

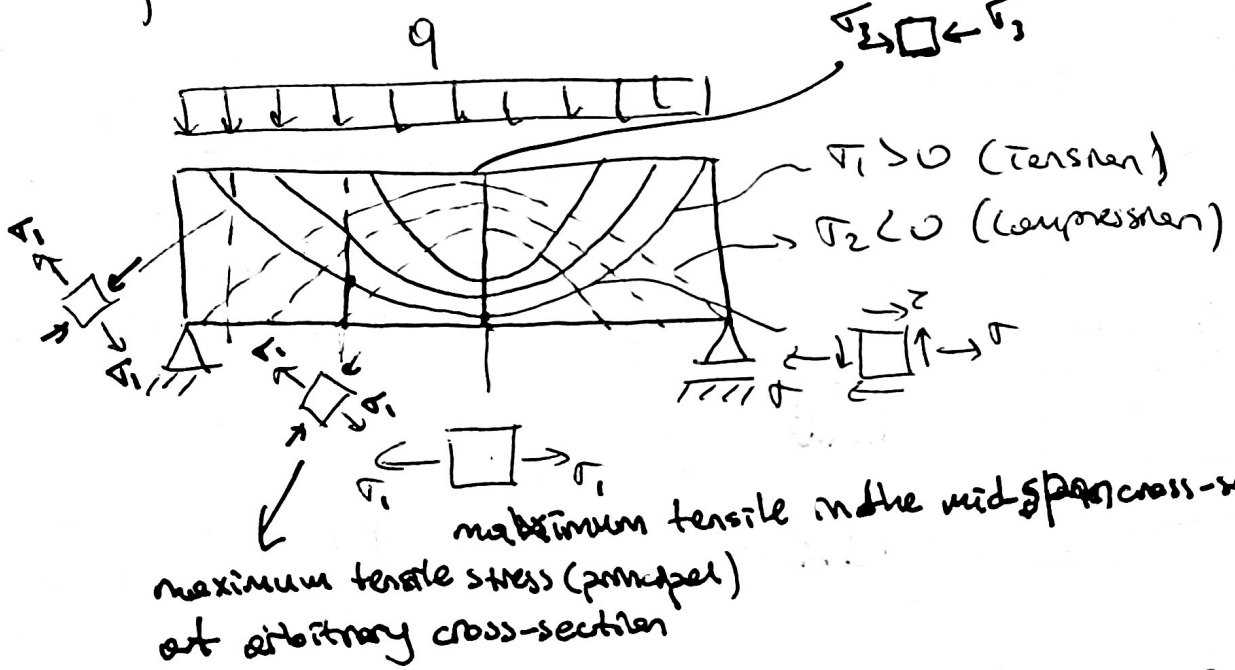
In two dimensional stress-state $\sigma_x = \sigma \quad \sigma_y = 0 \quad \tau_{xy} = \tau$

$$\sigma_{1,2} = \frac{\sigma_x + \sigma_y}{2} \pm \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$$

$$\sigma_{1,2} = \frac{\sigma}{2} \pm \sqrt{\frac{\sigma^2}{4} + \tau^2}$$

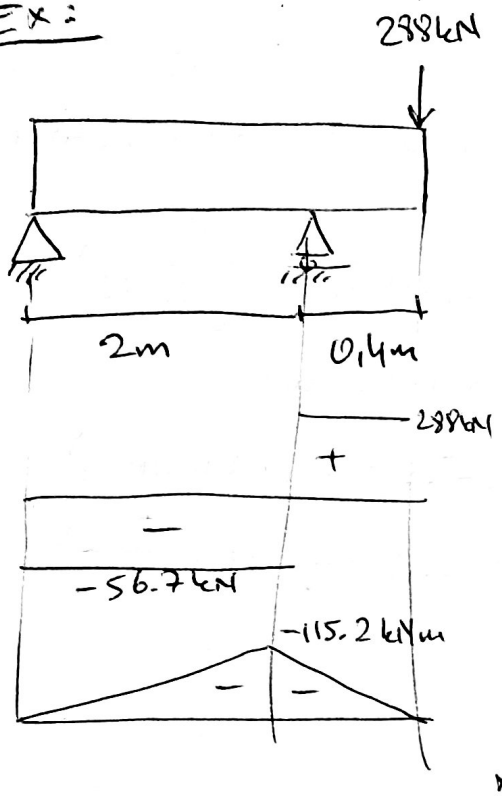
ORBITS OF PRINCIPAL STRESSES

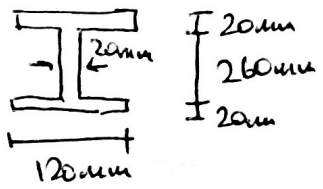
If the principal stresses (maximum tensile/compressive) are calculated at any location on the cross-section and the process is repeated for all cross-sections along the beam axes, orbit of principal stresses can be constructed



These trajectories give insight into the direction of maximum tensile stresses and can be used to determine the location of rebars for a concrete beam

Ex:





Check the safety of the cross-section with respect to

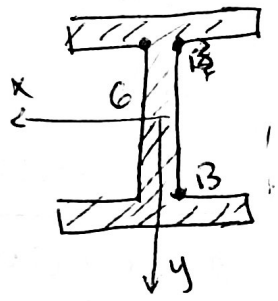
a) $\tau_{em} = 140\ MPa \geq \tau_{max}$

$\tau_{em} = 70\ MPa \geq \tau_{mm}$

b) TRESCA criterion (max. shear)

$$I_x = \frac{120\ 300^3}{12} - \frac{100\ 260^3}{12} = 12353\ 10^4\ mm^4$$

a) Max. shear stress occurs at gravity center. Hence



$$S_x = 120.20.140 + 130 \times 20.65 = +505\ 10^3\ mm^3$$

$$\tau_{max} = \frac{T_{max} S_x}{I_x . t} = \frac{288000\ 505000}{12353\ 10^4 . 20} = 58.8\ MPa \leq 70 \quad \checkmark$$

Max bending stress occurs at farthest point wrt neutral axis

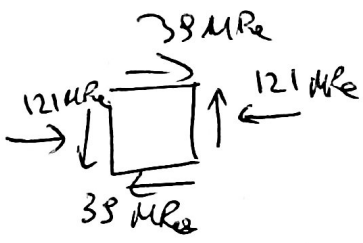
$$\sigma_{max} = \frac{M_{max}}{I_x} . y_{max} = \frac{-115.2\ 10^6 .}{12353\ 10^4} (-150) = 139.8\ MPa < 140 \quad \checkmark$$

b) The location between flange and the body are vulnerable (e.g B)

Shear and bending stresses must be determined.

$$S_B = 20.12.14 = 336\ 10^3\ mm^4$$

$$\tau_B = \frac{T_{max} S_b}{I_x\ t} = \frac{288\,000\ 336000}{12353\ 10^4\ 20} = 39\ MPa$$



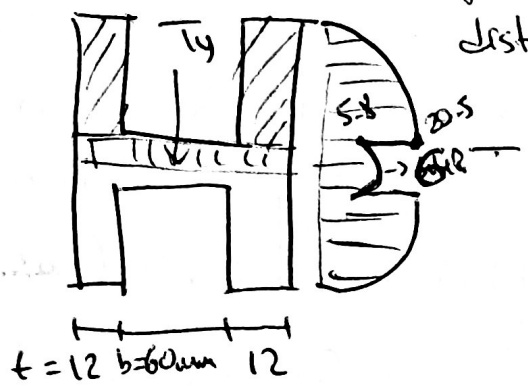
$$\sigma_B = \frac{M_{max}}{I_x} y_b = \frac{-115.2\ 10^6}{12353\ 10^4} 130 = -121\ MPa$$

$$\sigma_{1,2} = \frac{0 \mp 121}{2} \pm \sqrt{\left(\frac{-121}{2}\right)^2 + 39^2} \qquad \sigma_1 = 132.7\ MPa \qquad \sigma_2 = -11.6\ MPa$$

$\sigma_1 - \sigma_2 \leq \sigma_{em}$ ~~$132.7 - (-11.6) = 144.3 > \sigma_{em}$~~ 140 X

fail! $11.6 - (-132.7) = 144.3 > \sigma_{em} = 140$

Ex:



if $T_y = 40$ kN, then find the shear stress distribution and draw on the cross-section

$$I_x = \left(\frac{1}{12} 2.120^3\right).2 + \frac{1}{12} 60 12^3$$

$$= 3465000 \text{ mm}^4$$

$$S_{x_1} = 2.(54.12).-33 = -42768 \text{ mm}^3$$

$$S_x^2 = S_{x_1} + (6.84)(-3) = -44280 \text{ mm}^3$$

$$\tau_{zy} = -\frac{T_y S_x}{I_x t_s}$$

Vertical arms

$$\tau_{zy}\Big|_{y=6} = -\frac{T_y S_x'}{I_x\, 2t} = -\frac{40\,10^3 - 42768}{3465000\,(2\times12)} = 20.5 \text{ MPa}$$

Lateral arm

$$\tau_{zy}\Big|_{y=6} = -\frac{T_y S_{x_1}'}{I_x (b+2t)} = -\frac{40\,10^3(-42768)}{3465000 \times 84} = 5.8 \text{ MPa}$$

At the gravity center

$$\tau_{zy}\Big|_{y=0} = -\frac{T_y S_x^2}{I_x (b+2t} = -\frac{40\,10^3(-44280)}{3465000 \cdot 84} = 6 \text{ MPa}$$